SQL Server Exercises

EX 01 Create Stored Procedure

Create a stored procedure to insert employee records into the Employees table, then verify the inserted rows.

SQL

```sql
-- Create Tables
CREATE TABLE Departments
(
    DepartmentID INT PRIMARY KEY,
    DepartmentName VARCHAR(100)
);

CREATE TABLE Employees
(
    EmployeeID INT IDENTITY(1,1) PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    DepartmentID INT,
    Salary DECIMAL(10,2),
    JoinDate DATE
);

-- Sample Data
INSERT INTO Departments VALUES
(1,'HR'),(2,'Finance'),(3,'IT'),(4,'Marketing');

-- Create Stored Procedure
CREATE PROCEDURE sp_InsertEmployee
(
    @FirstName VARCHAR(50), @LastName VARCHAR(50),
    @DepartmentID INT, @Salary DECIMAL(10,2), @JoinDate DATE
)
AS
BEGIN
    INSERT INTO Employees (FirstName,LastName,DepartmentID,Salary,JoinDate)
    VALUES (@FirstName,@LastName,@DepartmentID,@Salary,@JoinDate);
END;

-- Execute Procedure
EXEC sp_InsertEmployee 'John','Doe',1,50000,'2025-01-15';
EXEC sp_InsertEmployee 'Jane','Smith',2,60000,'2025-02-10';

-- Verify Data
SELECT * FROM Employees;
```

OUTPUT

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
1	John	Doe	1	50000.00	2025-01-15
2	Jane	Smith	2	60000.00	2025-02-10

2 rows returned · SQL Server

EX 02 Ranking and Window Functions

Use ROW_NUMBER(), RANK() and DENSE_RANK() to rank products by price within each category, then return the top 3 products per category.

SQL

```
-- Create Products Table
CREATE TABLE Products
(
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(100),
    Category VARCHAR(100),
    Price DECIMAL(10,2)
);

-- Sample Data
INSERT INTO Products VALUES
(1,'Laptop','Electronics',80000),
(2,'Mobile','Electronics',60000),
(3,'Tablet','Electronics',40000),
(4,'Smart Watch','Electronics',20000),
(5,'Shoes','Fashion',3000),
(6,'Jacket','Fashion',5000),
(7,'Watch','Fashion',7000),
(8,'T-Shirt','Fashion',1500),
(9,'Sofa','Furniture',25000),
(10,'Dining Table','Furniture',35000),
(11,'Chair','Furniture',5000);

-- Top 3 Products in each Category
WITH ProductRanking AS
(
    SELECT ProductID, ProductName, Category, Price,
        ROW_NUMBER() OVER (PARTITION BY Category ORDER BY Price DESC) AS RowNumber
    FROM Products
)
SELECT * FROM ProductRanking WHERE RowNumber <= 3;
```

OUTPUT

ProductID	ProductName	Category	Price	RowNumber
1	Laptop	Electronics	80000.00	1
2	Mobile	Electronics	60000.00	2
3	Tablet	Electronics	40000.00	3
7	Watch	Fashion	7000.00	1
6	Jacket	Fashion	5000.00	2
5	Shoes	Fashion	3000.00	3

ProductID	ProductName	Category	Price	RowNumber
10	Dining Table	Furniture	35000.00	1
9	Sofa	Furniture	25000.00	2
11	Chair	Furniture	5000.00	3

9 rows returned · SQL Server

EX 03 Return Data From Stored Procedure

Create a stored procedure that accepts a DepartmentID and returns the total number of employees in that department.

SQL

```sql
-- Create Tables
CREATE TABLE Departments
(
    DepartmentID INT PRIMARY KEY,
    DepartmentName VARCHAR(100)
);

CREATE TABLE Employees
(
    EmployeeID INT IDENTITY(1,1) PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    DepartmentID INT,
    Salary DECIMAL(10,2),
    JoinDate DATE
);

-- Sample Data
INSERT INTO Departments VALUES (1,'HR'),(2,'Finance'),(3,'IT');

INSERT INTO Employees (FirstName,LastName,DepartmentID,Salary,JoinDate) VALUES
('John','Doe',1,50000,'2023-01-15'),
('Jane','Smith',1,60000,'2022-05-10'),
('Michael','Johnson',2,55000,'2021-03-20'),
('Emily','Davis',3,70000,'2020-07-30');

-- Create Procedure
CREATE PROCEDURE sp_GetEmployeeCount (@DepartmentID INT)
AS
BEGIN
    SELECT COUNT(*) AS TotalEmployees
    FROM Employees
    WHERE DepartmentID = @DepartmentID;
END;

-- Execute Procedure
EXEC sp_GetEmployeeCount 1;
```

OUTPUT

TotalEmployees
2

1 row returned · SQL Server